

硅谷101英文采访原稿

Interview with Nathan Lambert

Chen Qian: Hi Nathan, welcome to Silicon Valley 101.

Nathan Lambert: Hi, thanks for having me.

Chen Qian: You just came back from China, and during this trip, you visited many leading AI labs in China. Could you tell us more about this trip?

Nathan Lambert: When I came back, everyone asked me, “What was your biggest takeaway from this trip?” But I feel like a lot of information about who is building models and which teams are advancing is relatively public, so there weren’t many “mind-blowing” discoveries or feelings of “I have a bold new perspective.” For me, as someone who has been following these models for a long time, the trip was more about meeting the people behind them, understanding *why* they are doing it, and feeling the nuances. I’ve been trying to catch up on understanding China this year. In the US, many people try to explain for Chinese teams “why they are doing this,” but I wanted to hear it from them directly: “We are here, and this is why we are doing this.” That was very valuable.

So I’ll definitely continue following open-source models and trying them out. The ecosystem right now is fascinating. Many companies in the US want to build products and services around truly open model intelligence stacks to create value for people across different industries, and many of these models are actually built in China. These two tech circles, both with deep histories, haven’t interacted much in how they work. Some companies still have old rules—like Meta and Google—where if you travel to China for work, your laptop is basically unusable. There’s been a wall between us that made it hard to truly work together. But now this wall is slowly coming down. I’m very interested in that.

To some extent, it is surprising that these models are so competitive. But when you look deeper at the talent and the ecosystem there, it starts to make sense. A big reason I follow AI is because I think it will bring a lot of risks in the coming years, especially societal risks, like how people consume information. And of course, geopolitically, it will be a major issue. So building bridges between people and seeing the real humans behind these technologies is crucial, because they aren’t the ones making many of the decisions—even if the outside world views those decisions as highly politicized or extreme. Establishing these connections is really meaningful because this is a community working toward various goals—whether you call it AGI or something else. Meeting the people behind the technology was a real pleasure for me.

Chen Qian: That’s awesome. So how long were you there, and which cities did you visit mainly?

Nathan Lambert: About six or seven nights, mostly in Beijing and Hangzhou. We stayed in Beijing for three or four nights, right near the Olympic center, so I could go for a run around the Bird’s Nest in the mornings. The hotel was right there. But what’s really interesting is that most companies are located near Tsinghua University. There’s a whole cluster of AI startups there. It felt a lot like the Bay Area; you could just stroll around and hit all these top AI companies without having to worry too much about transportation. It really had that Bay Area vibe. Later, we went to Hangzhou for a few nights, where Alibaba is. I think DeepSeek is there too. We didn’t go to the DeepSeek office, but we talked to some people and visited Alibaba. It was great. This was my first time in China, and meeting these teams was amazing.

Chen Qian: So specifically, which labs or companies did you visit this time?

Nathan Lambert: I'll try to list them comprehensively. We also visited some robotics companies, though I didn't focus much on that part in my writing, just the language models. I went to two different Alibaba offices. As soon as I landed in Beijing, I went straight to Alibaba's Beijing office to meet the Qwen team. So there was Alibaba, Kimi (Moonshot AI), Zhipu AI, and Florian, who writes *Interconnects* with me, also went to Meituan, which has its own language model team. We also visited Tsinghua University and Xiaomi's campus in Beijing.

After getting to Hangzhou, we visited the Alibaba campus, the Bailing team at Ant Group, and Alibaba's ModelScope community. This was actually a group trip, largely organized around SAIL. SAIL is formed by a group of Substack writers in the AI space who have built a platform to do interesting things in AI media and events. So most of it was arranged around SAIL. Later, other members also chatted with MiniMax in Shanghai, and maybe StepFun. The list is really long. One obvious one is DeepSeek, which is the most mysterious, and also ByteDance. When you visit these companies, you realize almost everyone is talking about DeepSeek because they seem like a mysterious pioneer. ByteDance is that closed-source AI lab with massive resources that everyone seems a bit intimidated by. After all, their Doubao chatbot is mainstream now. You catch these signals quickly on the ground because everyone is talking about them.

Chen Qian: You wrote in your Substack article that you felt a sense of humility after coming back. What did you mean by that? As an American scientist...

Nathan Lambert: I don't feel qualified to comment on the Chinese tech ecosystem or the interaction between China and the US at a global level. I would say I knew I didn't know much before I went, and after going, I realized even more how much I haven't seen. I only visited two cities in China, and it was a very intensive work trip. A lot of the time it felt like being "teleported" by car from one place to another. Of course, we had great dinners, saw some lovely neighborhoods and old architecture in Beijing, but overall, I still feel I know very little about China.

If you want to form an immediate, accurate intuition about a place, you need a much closer full-stack knowledge and experience. What would someone growing up there think? What would be the instinctual reaction of a Chinese researcher? As an American, it might take me a lifetime to truly understand that. For example, what are the differences in the education system? When we went to Tsinghua, they also talked about an external impression—that in the past, Chinese researchers seemed less adept at "0 to 1" scientific innovation. That phrase itself is vague (what exactly is 0 to 1?), but they bring it up themselves. This involves how people are trained and how talent flows from broader domestic networks into different institutions. Hearing someone talk about it is one thing, but seeing it on the ground is another. This year, I read *Breakneck* and *Apple in China*, two great books about the interaction between the US tech ecosystem and China, but that's still a very limited perspective. So my feeling leaving China was: I knew I didn't know much, and now I'm even more certain that I have a lot to learn.

Chen Qian: Did you mainly meet with C-level executives, or mid-level AI researchers and engineers? Did you find these conversations helpful?

Nathan Lambert: We naturally wanted to talk more with researchers to establish a dialogue, but we also tried to meet leadership, especially leaders of startups. In practice, we mostly started by talking to researchers, but we also met some executives, like people from Alibaba, or people who are very well-known internally and deal directly with the founders. However, during formal visits, a big part is protocol, showing respect to the other party. But then again, company leaders are usually more media-trained, so sometimes they don't say much. They stick closer to company talking points, and details and nuances

get squeezed out. So I think it was a mix. Large companies might be more willing to let us meet senior cloud business leaders, but they wouldn't reveal too much. Some things were discussed privately, and I don't know—answers might not have been perfect, but it was a mixed bag. Especially when PR and comms teams are present; most formal meeting rooms have PR colleagues there. Ironically, some of the best conversations happened when I'd message someone, say "I'm in China," add them on WeChat, and then go grab boba tea privately with a researcher. Not in a company conference room. Then you can ask directly, "What are you working on right now?" without the superficial packaging of "I'm sitting in the office with my manager and PR next to me." Those conversations are more interesting, but rarer on this kind of trip.

Chen Qian: Exactly. So what questions did you mainly ask them?

Nathan Lambert: The questions had many layers. I'd start with a relatively easy one to answer: "Why are you releasing these models?" "Why build your own models?" "Why not just use DeepSeek's models?" Many companies are very pragmatic. They'd say, "We need AI to serve our users, and we want to control the full tech stack, so we build it ourselves." Compared to the atmosphere in the US where "building a frontier lab is super expensive and something only a few companies can do," Chinese companies aren't as daunted by building their own AI models. In the US, it might belong to a few elite companies, but in China, it seems like many companies are trying it. That's why we wanted to visit Meituan. We like them, but we didn't expect them to have their own large model, so we went to see what they were doing. They'd say, "Yes, we need this model to support our own agent products, so we built it. We are actually using it in products. Whether it's open-source or not isn't the most important thing, but we can release it to get some feedback." Their attitude towards "open science" is very pragmatic.

Following up on that, the conversation would naturally go to, "What is your biggest bottleneck right now?" Of course, this quickly hits the compute constraint. But many people would say Huawei chips are pretty good for inference. "We'll use Huawei for inference, but almost never for training"—many mentioned things like that. My peers on the trip would also ask more macro questions, like, "How do you think AI will develop next? How will it affect the labor market and the economy? As a scientist, do you feel a responsibility to understand and think about these impacts?" We found that Chinese researchers are relatively less willing to expand on their opinions. Part of it is corporate culture, but I think mostly it's because the environment they are in is much more confined than what is taught in schools; it's more about "I built the thing." Also, China doesn't have as strong a podcast ecosystem as the US, where researchers can gain recognition and even influence company direction through increased exposure. The overall vibe is much more pragmatic. Sometimes when writing, I mix this pragmatism with "humility," but actually, someone can be humble, while at the same time, they might just be in an environment where they aren't used to being asked certain questions.

Chen Qian: Okay, let's talk about some of the companies you visited. Did you feel they were very different from each other? Let's start with Moonshot AI.

Nathan Lambert: Yes, I think once you walk into these companies, they do feel very different. Kimi is an obvious example. You feel this group is incredibly close-knit. Building something cool gets them excited, and commercializing that cool thing excites them too. Their thought process is, "We are using the models we build ourselves." That closeness is something you instinctively feel the moment you walk into the office. You feel they have great cohesion, they are close, and they are fully dedicated to this together. So I would describe Kimi as having the "best vibe" among the companies we visited. You could feel they were happy and firmly on their own path. It didn't feel overly corporate. If you go to Alibaba, it's very obvious: this is a massive tech company, and the whole vibe is very corporate. Ant Group's Bailing

team has a bit of that too, as they also belong to a giant tech company. In companies somewhere in the middle, like Zhipu, the vibe is very AGI-oriented and very proud, but at least from what we saw, it didn't have that same affinity and charm as Moonshot. Of course, this could involve a lot of serendipity. After all, I only had lunch with these people for a few hours one day, and you can't judge a company's long-term trajectory just from that. But it's still interesting because people express their states, and you can definitely glean things from that.

Chen Qian: Moonshot has a paper called *Attention Residuals*, and one of the authors is a 17-year-old high school student. You know about this, right? Elon Musk retweeted the paper on X, saying "Impressive work from Kimi." This high school student is now famous in China. What do you think this says about Moonshot's culture?

Nathan Lambert: This might not be entirely unique to Moonshot, but rather that talent in these Chinese AI companies is very young. These companies are now trying hard to expand their presence in Western media, and many researchers are active on X. I knew some of them before. I feel like some of them are doing product work while also handling external communications for the company, like Crystal from Kimi, or Lu from Zhipu. I met Lu (I'm not sure if I'm pronouncing her name perfectly), but she's probably only 20 and still in school. When she said hi to me, I thought, "Wow, these people are really young." We also met a Xiaomi researcher, Lai Li, who I think is a PhD student but also one of Xiaomi's core researchers. Of course, the US has people like this too—people with a lot of drive who end up in frontier AI labs—but in China, it's just so common. You continuously meet these young researchers who are deeply involved in the core work of the labs at a very young age. This feels structurally a bit different. These companies absorb young talent much faster than many other labs, and they are more willing to let them step up. It's really shocking. I'm only in my early 30s, and I usually feel quite young, but these people are *really* young. It's cool to see though.

Chen Qian: But isn't it like that in the US?

Nathan Lambert: I don't think so. I feel...

Chen Qian: You mean there isn't as much young talent in US labs?

Nathan Lambert: Right. Ai2 has a feature, its secret sauce, which is that it's backed by the University of Washington. We have a lot of PhD students participating in OLMo and other major projects while studying. But this open approach isn't as accommodating in other labs. I know a few PhD students work in labs in the US, but they are usually the best of the best. In China, the integration with academia is much deeper, especially between Tsinghua and surrounding startups. This interaction is more intentional, and the overlap of talent is higher.

Why does this matter? Students are young, they have drive, and they don't have much else to worry about. They have fewer entanglements in life. I'm getting married this year and starting to think about a family. I'm 30; I have other things to care about. But a student thinks, "This is the tech I'm going to do for my whole life, I'm just going to keep doing it and master everything." When building language models, say if I do post-training, you need to know what the pre-training team is doing, understand the multiple stages of the training pipeline, and then find and fix bottlenecks. The technical knowledge you need is actually a very large, fixed body of knowledge. If this is the most important thing in your life and you have few distractions, it's easier to absorb all of it. One researcher told us that new students don't have preconceived notions about how deep learning used to be done. They don't start by wanting to change the architecture; they are more like, "I want to know what the state-of-the-art method is, and then I'll make it better." That is a simple, executable approach. Some Chinese researchers told us, "Yeah, this is what we are doing."

Chen Qian: I see. Let's talk about Zhipu. How was your experience there?

Nathan Lambert: It was very interesting. One thing I learned is that Chinese companies have a strong "showroom culture." When visitors come, they show the history of the company. In Zhipu's showroom, almost every display board had some AGI-related element, like "AGI loading..." or saying AGI progress is at 42%, which is a *Hitchhiker's Guide to the Galaxy* reference. You notice it as soon as you walk in. But on the other hand, I probably didn't get to know them as well as I got to know Kimi. However, they are very proud of their work and very successful. I feel that because they went public and were suddenly put on the US entity list, they were somewhat instantly pushed into a "truly big company" position. Kimi hasn't quite reached that stage yet. In my eyes, the two companies are actually very similar, just with different model trade-offs. In the future, if Kimi stays private and Zhipu goes public, whether the cultural gap will widen because of that is worth watching.

Chen Qian: Entering the secondary market does sometimes create differences. It usually brings more attention too.

Nathan Lambert: Yeah. Zhipu and MiniMax are both public now. It's good because they can raise money in the short term. But if they need more capital to train models later, it becomes harder—you can't exactly go do another IPO round. The situation is a bit unclear. Right now, the market is very enthusiastic about any public AI company, and investment demand is strong. But this trend could reverse. So it's a good thing now, but hard to say later. I don't make these decisions, just saying you can find reasons for both paths.

Chen Qian: Earlier this year, API requests for Kimi and Zhipu surged on OpenRouter when OpenClaw became popular. Do you think this trend will continue?

Nathan Lambert: Yes, we are seeing several different markets taking shape. At the very top are products like Claude Code and Codex, which I would call "knowledge work tools." But essentially, as long as a certain price point works, corresponding agent products will appear. The interesting thing about open models is that the price is low enough, making it easier for people to experiment. So naturally, many people try these models on this platform. These models are recognized as capable, they are a tier cheaper, and they have some brand recognition. DeepSeek, Zhipu, Kimi, and Qwen are all in this position. Personally, I think the name OpenClaw might be a fad, but the type of agent it represents will stay. For example, Hermes Agent is more popular now, but these specific products will come and go, and there will be new ones. The key is that you can run these agents directly, and calling these hosted models on OpenRouter isn't very expensive. Interestingly, OpenRouter itself doesn't account for a massive share of the real inference market. For example, in the US, Fireworks and Together AI are much larger, and they've recently started publishing data like how many tokens they process daily. But OpenRouter is more community-driven, so for people following open models, it's often the first platform they think of. This position is obviously advantageous. I think these companies are actively leveraging this momentum; when a model gets popular on OpenRouter, they push it. It's a great co-branding effect and self-reinforcing.

Chen Qian: Let's talk more about your experiences at Meituan and Xiaomi. This phenomenon is very interesting. Like we said earlier, the situations in China and the US are very different. In China, companies kind of like Uber and Apple are also developing their own open-source models, whereas in the US they aren't. Why is it like this in China?

Nathan Lambert: If you ask Meituan or look at the stance of companies like Meituan and Xiaomi, you feel their assessment is: "We will have many agents truly used by users in the future." Take Meituan for

example. You can imagine an agent doing things for a user, like helping them figure out what to request or what to order. It can do this.

Chen Qian: Yes. In the US, a consumer company's first reaction might be, "Then I'll just get an API key and call an external model."

Nathan Lambert: Exactly. But in China, many companies think, "We can do this ourselves, and maybe it will be cheaper. We have the compute and the talent, why not try doing it ourselves?" And it's not just Meituan; several companies think this way. "We build a general model, but our use goes beyond that. The general model can be released for more people to use." Then they fine-tune it further into specialized agents and specific models for their internal services. Meituan is a great example. They won't release those specialized models, but they hope to put them into their apps. You can imagine Xiaomi, for example, the model could drive certain features in their cars. I guess Xiaomi might act like Tesla, where cars communicate with the cloud via various interfaces. Though I'm not sure what it will look like exactly, AI will increasingly integrate deeply into all digital technologies. I think Xiaomi is also a typical case of "a big company wanting to do everything themselves." Now they are doing large language models, robotics models, and robotics hardware. They really want to do everything from scratch themselves.

Chen Qian: Yes, Xiaomi's models are actually quite new. From April to June 2025, they released their initial MiMo 7B model. By December, they launched the very popular MiMo-V2-Flash, and recently I think V2.5 came out. It takes a little time for people to realize that these models will keep emerging. Seeing a new lab build up almost from scratch is indeed surprising. They also hired former DeepSeek employees... I forget how to pronounce her name.

Nathan Lambert: Luo Fuli.

Chen Qian: Right, Luo Fuli. Now it looks very likely that Xiaomi will become one of China's strongest modeling teams. But in the US, many people's knowledge of Chinese models is still stuck on DeepSeek, Qwen, Kimi, and Zhipu. Some know MiniMax, some might not. The list is actually already very long. But after visiting Xiaomi, my hunch is they will keep at it. They come across as very pragmatic, and to them, doing this makes sense. Xiaomi has already accomplished many impressive things in the past. Compared to building cars and other hardware, building an AI model suddenly doesn't seem that crazy. Meituan also invested in Moonshot's recent funding round, which shows Meituan is placing big bets on AI. Chinese companies often have massive ecosystems. They love super apps, they love ecosystems, and they love building everything from scratch themselves. Do you think this is the right path in the AI era?

Nathan Lambert: It is definitely worth trying it hands-on. I think because AI is inherently very resource-dependent, it favors large companies, especially those that can enter a positive flywheel: "We have a lot of resources. AI makes our products better, so we get more returns, which we reinvest, and ultimately we win more." This trend is already obvious in the US. Google is in a very strong position. OpenAI and Anthropic essentially have nothing to worry about. Nvidia, needless to say. Any company in this feedback loop will be very strong. It's interesting to see how this logic will unfold in China. I think Ant Group's Bailing is on this path. They are building their own models while seeing their platform capabilities expand, for example, they are entering healthcare now. They talk about this often, and will definitely apply models to these scenarios in the future. So for them, this makes total sense. But if you are a Western observer, you might wonder why Ant Bailing even exists. They don't realize how massive the business footprint and application scenarios are behind a platform like Alipay. So I think for these companies with massive reach, building their own models makes sense. However, I also worry this

might suppress some competition. For instance, how do smaller companies like Kimi, MiniMax, or Zhipu fit into an ecosystem where massive applications already exist? I can't remember which company it was, but they told us they built an agent to operate an app owned by ByteDance, and ByteDance blocked it. So they couldn't continue scraping data. You can see that between big and small companies, new interfaces and boundaries are forming, or rather, small companies are striving to find their niche. Meituan is somewhat in the middle here. From my perspective, I still haven't fully grasped exactly how big a tech company it is considered, whether it will be swallowed by industry consolidation, or if the cost of continuing to invest in language models will become too high for them. I'm not sure about these things. But overall, I'm still very optimistic about AI as a business, so I think these early entrants will likely do fine. They are selling tokens, selling intelligence, and there will be massive market demand for these things.

(Note: The transcript is quite long. This covers the first ~25 minutes of the dialogue. Continuing...)

Chen Qian: You wrote that Chinese companies have a "technology ownership mentality." Can you explain that?

Nathan Lambert: This ties back to the super app idea. They just feel they can build the thing, and build it better. If successful, they keep investing money, keep improving efficiency, and tune it for their specific scenarios. This is very similar to the path many Chinese domestic industries took to become some of the largest global industries: do it yourself if you can, rather than buying someone else's service. That's how you control the long-term trajectory. It's like—I think it was Wang Dan in *Breakneck* who said that Chinese capitalism is more intense than America's, because they think over longer cycles about what will bring the best profit margins, the strongest product capabilities, and competitive advantages. If you rely on someone else's service, it's hard to truly beat them. In a highly competitive environment, you want to do everything best, and ideally, do it yourself. So I understand this as a "keep it in our own hands" mentality. For these companies, building a model isn't yet an entirely unaffordable expense. I estimate they might spend \$500 million to a few billion dollars a year on these models. It's a lot of money, of course, but it hasn't reached the point where these labs can't afford it. If model capabilities continue to scale up, the market will likely eventually consolidate, because capability improvements look somewhat logarithmic to investment—you invest more, the model gets stronger, but with diminishing marginal returns. Right now is a very interesting phase. I remember interviewing Richard from Ant Bailing; he said it only took them six months to build the team making large models. This is because they already had a massive talent pool; they just reallocated a team to do this. So we are in a phase where I don't know how long these companies will keep investing, but at least for now, it's very interesting.

Chen Qian: Let's talk about the other companies you visited. Qwen and Alibaba are big companies. They are different from Moonshot and other smaller labs, right?

Nathan Lambert: Right, I think Qwen won broad recognition from the open-source AI community by continuously releasing highly capable small models covering a wide range of sizes. They were actually doing this before Llama 4 came out and lost momentum. And when Llama 4 was released and Meta slowed its pace, Qwen really hit its stride. They know this too. So when we asked them, they were very focused on what developers would use. They want to continuously build models that developers will use and build upon. But if you look at Qwen, you'll notice they haven't released their largest models for a while. For example, Qwen Plus and Qwen Max are closed-source. This clearly reflects the Alibaba Cloud side of things. Just like Google Cloud in the US, there are huge expenses and huge revenues. In China, Alibaba is the most natural company to take this position, and they know it. So they won't necessarily release all models. Maybe they'd say they don't release them because no one would use

them, but I also think, "There is a lot of money to be made here." And there really is a lot of money in AI. The most interesting thing is that many people mentioned their focus on small models is intentional. It's not an accident or just "others are doing it, let's try." They have a very mature recipe for models at the 1B, 10B, and 30B scales. These models are very popular, and it's by design. Hearing them say, "Yes, this is how we do it, and we know it takes a lot of time"—releasing a model is hard; it requires a lot of work for the ecosystem. I think they also feel the pressure: "We could make more models, but whenever we release one, someone has to maintain it. We have to interface with vLLM, with all these companies, and integrate into the open-source ecosystem." The model teams themselves are still small and can't outsource this work; it has to be done by those directly involved in developing the model. Thus, there is caution on both sides. On one hand, they can't release too many models; on the other hand, they might not release the absolute largest model because they want to monetize it on the cloud.

Before this trip to China, another question I cared about was: "Is AI investment more like SaaS software spending, or more like cloud spending?" In the US, there's a view—whether a stereotype or reality—that Chinese companies are less willing to pay for software services. For example, the equivalent of Slack is much smaller. Many companies might rebuild an MVP themselves or download open-source tools rather than pay a per-seat monthly software fee. But these companies all use the cloud. Alibaba Cloud itself is a massive business, and modern software must run via the cloud. I think China will increasingly view AI spending as cloud spending. It might not be as extreme as in the US, but it will be seen as a necessary input to build the technology they want. Therefore, these cloud providers will exist. It's very interesting because once you arrive at Shanghai airport or take the bullet train to Hangzhou, you see Alibaba Cloud billboards everywhere.

Chen Qian: Yes, as soon as I arrived in Beijing, the first thing I saw was an Alibaba Cloud ad, thinking, "Okay, I'm here." I think one difference between Alibaba Cloud and AWS is that AWS Bedrock hosts almost all large models, proprietary and small models. But Chinese cloud providers usually only host their own models. That's a big difference. And I think the US has a new type of cloud service market where many people resell others' models, charging per token, or reselling compute. This is a big chunk of the US market. You can list many companies: CoreWeave, Nebius, Together, Fireworks, Lambda, etc. This layer is much more open. We also talked about OpenRouter. I don't know if China has a true equivalent.

Nathan Lambert: I don't think so. In the US, many of these companies are seen as being backed by Nvidia in many ways. Nvidia wants the ecosystem to be more diverse, because if most revenue only comes from a few companies like OpenAI, that's bad for them. China doesn't seem to have the exact same dynamic.

Chen Qian: Yes, this is very interesting because China is missing this layer. They don't have this infrastructure or optimization layer. If you look at current trends, like Cursor taking Kimi's model, doing post-training at Fireworks using their RL sandbox training environment, and then serving it through Fireworks—that's a very interesting multi-party collaboration. Several companies working together to make strong models and products. If you wanted to do the same in China, it might require more vertical integration. Maybe Chinese companies just prefer to do everything themselves, keeping it in their own hands. This runs contrary to the logic of "ownership" and "integration." To support a company in the middle layer, you have to be willing to hand over a critical component, whether it's GPU hosting or inference serving, because you want to focus your energy elsewhere. This looks more like the Western startup model. However, I do think many Chinese model labs, compared to older, larger Chinese tech companies, will be more willing to try different approaches. Kimi knows Cursor is training on their models and will go talk to Cursor.

Chen Qian: 01.AI is the AI company founded by Kai-Fu Lee. It has now undergone a major transition, pivoting to a more B2B commercial model. How do you see the future of the B2B market in the Chinese AI era?

Nathan Lambert: This comes back to whether these partnerships can form. I think many companies are wondering if they can enter the global B2B enterprise software market. Honestly, it's hard for me to judge if the domestic B2B market will ultimately win. You can certainly bring AI expertise into another company, which is very valuable for companies with weak AI capabilities. But the question is, are people willing to do that? Because once you rely on an external partner for this, you lose some control. The recent trend in the US—FDE (Forward Deployed Engineer)—is a buzzword in this space. OpenAI is hiring heavily for this. It will happen in the US, but will it become a viable model in China? Can you walk through the front door of a company and be welcomed? It's very interesting. They will try. I think if this path works, the first person to successfully pull this off in the AI era will obviously be very valuable. But I can't say for sure. It's a coin toss, 50/50. It just depends on Chinese culture and the business environment, which I can't completely understand yet.

Chen Qian: Did you meet anyone? We met Kai-Fu Lee. How was that conversation?

Nathan Lambert: Good. Meeting him is like meeting a legend. It's really great to meet someone like that. He is very focused. He has already done so much. Theoretically, he could do whatever he wants, even not work. But you see him still trying to solve problems, reimagining how this business should be run. This fits the ethos of the AI era perfectly. AI has drawn many such people back, making them rethink how the next phase of companies will be built.

Chen Qian: Why didn't Kai-Fu Lee's lab become China's leading lab?

Nathan Lambert: I really don't know. You should ask him. It's a tough question.

Chen Qian: Yes, I think there are many opportunities in the AI field, but building models is a completely different culture.

Nathan Lambert: Building models is like constantly cranking a machine and continually doubling down on resources. I don't think 01.AI's situation is simply "they built one model and stopped." There are similar situations in the US. For example, Databricks built models, and later they might have felt, "We don't need to keep doing this; others have already figured it out." While I'm not sure if that's their public stance, I imagine much of Databricks' management thinks, "We are glad we don't have to spend money training models anymore. We will focus on our core business." This is also why I'm surprised that so many companies are still doubling down, wanting to build trillion-parameter MoEs, which is much more expensive. 01.AI and Databricks exited at the small MoE or small dense model stage, where the investment is much smaller. From a certain angle, if you judge that you won't be competitive, stopping early might actually be smart. Not everyone needs to own or build the best model, right?

Chen Qian: Right. A classic but important question: how many months do you think top US models lead Chinese models? Will this gap narrow or widen? Why?

Nathan Lambert: On traditional benchmarks, it used to be around 6 to 9 months for a while. But in real-world usage, especially in the early days of reasoning models, the gap was very close because it was a new paradigm. In a new paradigm, the returns on improvement are very steep. Both US and Chinese labs were climbing very fast, and Chinese labs release faster. If you talk to people at Chinese labs, they'll say, "Our model is still undergoing RL. The final RL round takes about a week, and then we release it within a day of completion." US labs usually have a process; after the model is done, it takes

weeks to release. So this is rapid growth with two different curves. If China is in a trailing position, it's not at zero progress. The horizontal line phase is the waiting period before release. Since China releases faster, they effectively narrow the performance gap users experience. Ultimately, the performance gap is the difference between the models people can actually use.

We discussed the knowledge work domain earlier, which mostly involves American workers. I think American frontier models are more robust in their raw utility for these real-world scenarios. But this is hard to measure. So on popular benchmarks, the 6 to 9 month gap might remain relatively stable, but in intangible, hard-to-measure usage aspects, US models might pull away. The best example is the programming revolution that started in December/January with Claude Code and Codex. If an open-weight model could support the same behavior, the whole world would talk about it. If I pay \$1 a month instead of \$200 to OpenAI, many people will switch. Right now, I'd say the gap is five or six months. I don't expect a very clear watershed moment this year that will make us redefine the gap. The way we view the gap now might be too narrow for the scenarios where AI is widely used today.

Chen Qian: Why hasn't China produced a truly excellent AI coding model yet?

Nathan Lambert: A big reason Claude Code and Codex are popular is that they are extremely easy to use. I think Chinese teams will continuously solve the coding capability issue. The actual code quality of the models will be very high, because this can be learned by having the model solve problems using public data, GitHub, etc. But on "ease of use," US labs have a big lead. The operating model is also different. US labs can get consumer data. For instance, Cursor trains models based on users using Cursor. ChatGPT, Codex, and Claude Code are similar. When open-weight labs release models, although they have their own CLIs, they aren't as popular. Massive model usage doesn't flow back to the developers. This real-world data is incredibly useful for improving a model's robustness across various general scenarios. This is a huge advantage for closed-source labs.

Chen Qian: Okay, I have a few more questions about open-source models. Why do Chinese labs release so many powerful open-source models?

Nathan Lambert: It's their path to relevance. I think they know they are behind, and they want to be part of the conversation. If no one knows you, even if you have an API, no one will think of your model. But in a world where AI models are extremely valuable and relatively scarce, when you open up a model, many people say, "I want my own AI model too," and they come to look and try it. So these labs are very pragmatic. "This is how we participate in the frontier AI openness story." They will continue doing this. I expect this to be very stable, because the fast-follower gap will persist. It's a very pragmatic mindset: "Doing this doesn't add any risk, it helps us achieve our goals, and it's one of the only ways to get customers. So we have to do it."

Chen Qian: If Meta fails to become the best open model, does that mean the US is losing open model leadership to China?

Nathan Lambert: Over the past year, the US has already ceded open model leadership to China, especially starting in the summer of 2025. A new crop of Chinese labs became very prominent. GLM 4.5 and Kimi K2 emerged as breakthrough models. Since then, the US no longer holds open model leadership. I think it's been this way since summer 2025. Anyone claiming otherwise is a bit blind to where the truly important models are coming from.

Chen Qian: Does the US still have a chance to catch up?

Nathan Lambert: Yes, because the US has Nvidia. And Nvidia's financial interest is that AI shouldn't be owned by any single company. The best way to achieve this is to release open models for the whole world.

Chen Qian: So Nvidia is the only hope?

Nathan Lambert: Realistically, probably. On the open model front, companies like Meta and Microsoft have their own products, whether for consumers or enterprises, supported by AI. So they could release open-weight models without truly hurting their business. But over the last few years, they've chosen not to. Maybe it's a distraction, maybe there are marketing or security risks. As long as this remains the case, Nvidia is in the best position to change the game. Just days before we recorded this episode, Jensen Huang released their Nemotron 3 Ultra model. I think this means Nvidia has completed the first phase of retrying open-source models. They launched the Nemotron 3 Nano, Super, and Ultra series, using an MoE architecture, with Ultra being the largest. Next, they will use it to generate synthetic data, building various small models and distillations internally. Now that they have their first large model, they have more flexibility to build other sizes and keep improving. So Nvidia's moves in open-source models will be worth watching closely. The question is, once someone builds this large MoE model, can they continue scaling it up from there? Nvidia has had many models in the past with decent quality, but they quickly fell out of the usage and discussion cycle.

Chen Qian: What about players like Reflection AI or your team?

Nathan Lambert: I think more people will build models in the open-source community in the future. The open-source ecosystem itself can accommodate various types of models. These small and medium models are incredibly useful because they are so cheap. If you can have it repeat a task, you save a lot of money compared to constantly using Claude or Codex. There is an obsession in the open model space right now with chasing the frontier, chasing the biggest MoE models like Nemotron Ultra or Kimi K2. I think over time, this obsession with the biggest models will fade. Reflection AI wants to build a large model, but they have to compete with Qwen. And Qwen has already won recognition. The model scale isn't huge, but every researcher around the world uses it as a foundation. It's unbelievable how much AI research is done on Qwen now. For a long time, it was Llama, but now everyone, every researcher knows exactly how Qwen works and is used to it. This is an incredible achievement for Qwen. I feel it's not discussed enough. Over time, interest will be distributed across more layers, from small to large models. For Reflection AI, maybe picking an underserved niche is better. Ai2 builds small OLMo models, which are extremely helpful for research. I don't think Ai2 should chase the frontier and build a trillion-parameter model; from a philanthropic funding perspective, it's not feasible. But building something strategically different is also good. Before people realize they can't financially keep up with the frontier, there is still time to fill different niches.

Chen Qian: Whose open-source model is the best in China?

Nathan Lambert: The ranking changes back and forth. I'd like to spend more time trying out various models myself. If you ask me right now what the strongest models are, I'd say probably Kimi and Zhipu. For instance, Kimi K2.6 and GLM-5.1 are the ones I hear about most. They seem to genuinely make an impact in frontier real-world use. But it changes so fast. For all of 2025, DeepSeek was basically the undisputed king, especially with the V3 and R1 series. I wouldn't be surprised if Xiaomi catches up from behind and takes the lead. In China, Qwen's largest models have always been closed-source, which to some extent limited its adoption and buzz on larger models. But I can't say for sure. If you point a gun at my head and force me to say, I would say Kimi and Zhipu are the best open-source models in the world.

Chen Qian: But the competition is fierce, right? Always changing. Very slim margins. You mentioned many Chinese researchers in your article. Can you talk about your impression of them? How are they different from US AI researchers?

Nathan Lambert: China has a tremendous amount of talent. When you go into a lab, you'll find mostly home-grown researchers, and they are often younger. We went to many places near Tsinghua. Of course, China has other great universities, so when you enter a Chinese lab, the demographic composition is more homogeneous. By contrast, at Ai2 or other US labs, as an American who isn't an immigrant and whose parents aren't immigrants, I sometimes feel a bit unusual. I work with people from all over the world: Europe, the Middle East, China, other parts of Asia. US labs are like this; there are many Chinese researchers, but looking at the overall demographic, the vibe is very different.

Another difference is what we mentioned earlier: when top US researchers discuss "where AI is going," they often center on the philosophy behind the models or artistic design concepts. An example brought up by Caithrin Rintoul, an organizer of this trip, is someone like Amanda Askell talking about model persona and how to meticulously shape a model's tone. Such discussions didn't really happen in China. There are some local discussions in China, but the topics they are willing to discuss and how far they are willing to project AI's future are very different from the US. Other than that, they are actually very similar. An interesting and uniquely Chinese point is that the English of the newest generation of Chinese researchers seems much better. Many very young people speak excellent English and are paying attention to the Western AI ecosystem. They are on Twitter, reading blogs, and writing papers. But the reverse isn't true. In the US ecosystem, many people don't care what happens in China and don't really understand how the Chinese tech ecosystem operates. Even for someone like me, who is very interested and wants to know how these companies do things, getting reliable information is hard. So the information flow is very asymmetric, which affects how people think.

Also, there is a very typical American trait of being extremely proud of what is happening around them, even a bit narcissistic. Silicon Valley especially rewards this expression. Silicon Valley is developing "tunnel vision," forming increasingly narrow culture and discussions around AI, like weird narratives about a "permanent underclass," or talk about recursive self-improvement during the technological singularity. You hear a little bit of this in Chinese labs, but much less. They might receive some of it from the US and think it's a bit strange, though interesting to listen to. But if you go to San Francisco, people there are genuinely talking about this. You'll see PhD students saying, "I need to make money fast, I need to get into a lab." There is a very intense culture and anxiety surrounding AI timelines and career changes. It feels different in China. They are more like, "We are going to build this thing." In a sense, they are more grounded and less anxious.

Chen Qian: How do you view Chinese AI talent in this wave? Even looking at the leading frontier AI labs in the US, you see many Chinese researchers, right? Many Chinese researchers enter these labs after finishing their PhDs in the US, or they might be immigrants or second-generation immigrants. What role does Chinese talent actually play in AI research today?

Nathan Lambert: They play a very critical role in every company. I think the US's current political stance on restricting high-skilled immigration is detrimental to the US itself. I work with many Chinese people, as well as people from other international backgrounds. I want them to be able to stay in this country that I also love living in. So I disagree with these policies being pushed. And I think anyone working in the tech industry knows how critical this talent is. Of course, there are concerns. For instance, recent news about executives of some AI companies being restricted from leaving the country. Some people in the US know how important talent is, and with tensions rising, they worry, "What if they recall top researchers?" Because if that happens, US companies would take a severe hit. That's the

reality. I don't know if anyone will actually make that decision in the short term. Decisions like that are very far removed from the audience of this podcast; the listeners aren't the ones making them. But in the US, people do discuss this as a major potential bottleneck for the tech industry's future development.

Chen Qian: It sounds like as long as you have one person to lead a team, you can build a model, right? And then you can build it.

Nathan Lambert: I think vision itself is hard. Building language models involves an often underestimated part: it is largely an organizational architecture problem. You have a lot of talent and you want to make the model better. The really hard part is how to organize these people, how to break a big problem into many small ones so that a massive amount of talent can contribute to improving the same model. I'm not sure I want to state this absolutely, but one possibility is that the way Chinese engineers are trained is indeed better suited for this organizational model. Everyone focuses on polishing their small piece perfectly, accepting that it ultimately contributes a part to the larger model. Of course, there's a lot of uncertainty here because how frontier labs actually operate isn't well-documented. Maybe reality is messier—just a bunch of people running around trying to figure out how to make the model better. The analogy might not be perfect, but it's certain that this is heavily bottlenecked by organizational capability. You must organize so many people effectively towards the same goal.

Chen Qian: Are Chinese AI researchers more willing to drop old assumptions and directly absorb the latest paradigms? Like MoE, RL agents, coding models, and tool use. MoE wasn't invented by DeepSeek, but DeepSeek optimized this technology to an extremely efficient degree.

Nathan Lambert: I think this has more to do with young and motivated researchers. There's a depressing saying: "Science advances one funeral at a time." That's probably the most pessimistic way to put it, but the core meaning is that newcomers are more willing to embrace the latest things. They don't have past achievements to maintain or old viewpoints to defend, so it's easier to jump right in. To some extent, Chinese engineers and leadership might be younger overall, with less experiential baggage. So this idea might hold true. If you do research, you know many new ideas and changes indeed come from young people.

Chen Qian: From the perspective of Chinese culture, is it harder to cultivate truly visionary people, top experts at the level of Ilya Sutskever?

Nathan Lambert: I do have that impression. And that impression is based on the culture as I understand it. That is, from a very young age, the whole system is built on standardized testing. Some of these tests reward memorization. Students know clearly that if they do well on this test, they can get into the best school for the next stage. So they are constantly in a highly competitive state, focused on this critical threshold that decides their future. Please correct me if my understanding of the Chinese system is flawed, but my understanding is that in the US education ecosystem, these "one-checkpoint-decides-your-fate" nodes are relatively fewer. It's more of a continuum. Many of the best people go to the best schools, but tier-two schools are also very good, and there's more crossover and mobility between tiers. The US certainly has credentialism and school prestige, but maybe not as extreme. So this is very likely true. Some leaders will privately mention that the education system rewards memorization, and even if you wanted to change it, reshaping the education system is incredibly hard. When some tech leaders say this privately, I think, "Okay, I have no reason not to believe you."

Chen Qian: You described Chinese labs as near-perfect "fast followers." Can you explain what makes them execute so fast?

Nathan Lambert: This ties back to what I mentioned earlier about young talent and culture. Very tech-oriented, very task-focused. People might lock onto one thing and just keep doing it. And they don't lack talent; they can cover the entire tech stack and eventually converge on one model. We've seen this many times. Once you know something is possible, building it becomes easier. Chinese tech companies have shown this capability in other fields too. For instance, if you go to an offline Xiaomi store, it feels a lot like an Apple Store. Obviously, they were inspired by the iPad, iPhone, and Apple design. And they indeed do it well. They haven't just done this in AI. I think they will do it again in AI. Right now, the debate is whether doing these foundational things and polishing them repeatedly is enough to let them... Everyone is guessing, "Will China surpass US AI labs?" I think if resources were completely equal, they probably could, or maybe they just have more talent willing to put in the time over five to ten years. But I think talent is only part of the resources needed to build models, while compute and data capital are the biggest levers. Compared to US hyper-scale tech companies, Chinese teams seem to have significantly fewer resources. So talent alone can only take you so far. However, technology diffuses. Once the best model appears, making a more efficient model is much easier. So it will keep advancing like this.

Chen Qian: Is it possible for an adept follower to become a leader in frontier fields?

Nathan Lambert: Yes. AI is special because the resource requirements are so massive. China's path to leadership might not be immediate overtaking, but because the timeline to the ultimate goal is long enough. And during this time, Huawei builds up production capacity, net compute capacity grows... so when people discuss the balance of power and chips, you often hear this scenario. Most people think it will take a long time, but you can't rule it out.

Chen Qian: We just talked about the cultural differences among top AI scientists, how it is in the US. Back in China, maybe Xiaomi is an exception. For instance, Luo Fuli is quite active on social media and podcasts. And Lin Junyang from Qwen, who was Qwen's technical lead but left the company earlier this year. Especially with the Lin Junyang case, I feel you can see a clear tension between individualism and collectivism/company-first interests. Do you think we will see more of these conflicts in Chinese labs?

Nathan Lambert: Entering the Western ecosystem, I think some Chinese researchers will find they are actually very suited for this environment. I'll call him Justin, that's his English name. He is really good at this. I see it as a phenomenon we've also seen in the US: some people in tech companies who have strong external influence might end up challenging company management. But big tech companies often feel, "We don't necessarily need to listen to you," so the two sides might part ways. Similar things have happened, like with Yann LeCun. The US has seen this, a classic example being some AI ethics researchers at Google. Back then, Timnit gave Google management something like an ultimatum, and the company rejected it. In other words, inside big tech companies, some walls are very hard. I think some Chinese researchers might also enter this ecosystem, and we might see similar situations again. But if it's in a startup, like Kimi or Zhipu, I would actually welcome someone who can bring buzz and attention. The rest of the problems can be solved later. This largely depends on company size, but I think it would be a good thing if more Chinese researchers enter this public expression ecosystem, because then we get to see more perspectives from another part of the world. If they can be better understood by more people in the US, everyone can better understand how they think and build models. This is beneficial for global balance.

Chen Qian: That's great. We are trying to do this too, inviting more Chinese researchers onto the show. So everyone can subscribe to our channel. When you chat with Chinese researchers, what is the vibe inside Chinese AI labs like? Do you see more confidence, anxiety, curiosity, or pragmatism?

Nathan Lambert: I think they do have a kind of anxiety, mainly worrying about compute not keeping up. I think all Chinese labs want more Nvidia compute to train the newest models. And researchers are the ones who feel this most directly, because they are the ones constrained by resources. But other than that, they have less societal anxiety. In the US, social anxiety is already very intense. For example, colleagues around me genuinely worry about their jobs being replaced. I'll think, "You are already one of the most brilliant AI researchers I know." But even people like that will say, "I am really, really worried about this." This anxiety also appears in broader public discourse. US CEOs go around saying AI will put many people out of work. I don't think that's a good way to communicate, but it has become deeply embedded in how people view AI now, and embedded in discussions about data centers. But these cultural focal points don't seem to have really landed in China. When some developers are asked, "What is AGI?", they say, "It's when it takes my job." But it's not like a dark cloud hanging over their heads. They are more like, "I want to build this thing first and see." Why is this? There is some hesitation in the developer community too, and things change fast. They are all using Claude. Even though Claude is theoretically inaccessible in China, they use it for work, and they tell us honestly, "We love Claude. Claude is the best model." So there is some growing unease about software jobs changing, but it is far from forming a collective social judgment that "AI is bad and will destroy society." In China, this narrative doesn't seem established. I think some people also mentioned that China has experienced so much rapid technological and economic change over the past few decades, so they feel AI is just the next wave of change. They've been through many rounds before, like China's super apps, and now EVs. People are used to it. In the US, the intervals between these changes are long, the pace isn't as dense. So when the newest wave hits, it faces more inertia and resistance.

Chen Qian: Do Chinese labs internally talk about alignment or safety?

Nathan Lambert: They more commonly use the word "security." Companies obviously care about these things, but it's not their highest priority. Their thinking is more like: if we release model weights, we must ensure it doesn't bring some massive new risk. We would follow up with questions, like, "What about multimodal models?" They admit some things might become risks faster, but they don't have that intense narrative of, "The text LLM agent we are releasing is a major risk, therefore it must be banned." Their answers to these questions are reasonable, but unlike in the US. In the US, this discussion sometimes takes on an almost ideological tone. It's also a very deep concern that the *next* generation of AI might be the one truly worthy of high alert. So I think Chinese labs have a measured attitude, but this issue isn't put at the forefront. It's possible Chinese companies are just more pragmatic. I think that makes a lot of sense. To some extent, this is close to my position too. I support open models, but I'm not an absolutist. I don't think every single model should definitely be open. For example, Claude Mythos—to some extent it involves marketing packaging, of course, but it is indeed a very major breakthrough. If it were uploaded to Hugging Face on day one, that might not be a good thing. Technology diffusion naturally takes time, especially for technology that could have massive impacts. Gradual diffusion over time is reasonable. In the current state where closed-source models lead and open-source models follow, I actually think this is a better way, allowing technology to step-by-step enter the real world. I think many Chinese companies probably feel similarly.

Chen Qian: In what aspects do you think Chinese labs truly lead US labs today, and in what aspects are they just moving faster than Americans expected but not truly leading?

Nathan Lambert: Many people say Chinese labs have built more efficient models. I think this claim is slightly overstated, because US labs handle a much larger total volume of inference. They have much stronger financial incentives to improve efficiency. For example, increasing Claude or ChatGPT's inference efficiency by 1% could mean billions of dollars. So US companies actually have every reason to pursue efficiency. But Chinese labs, because they have been releasing open-source models for

longer, genuinely understand better how people use these open models. When we talked to Alibaba people, they were very consciously doing this. The same goes for Kimi, Zhipu, and other teams that have been through multiple MoE releases. They are tracking how users use models. This kind of specialized knowledge is very fleeting and hard to acquire. I've tried to figure out how people use OLMo and found it very complicated; direct information is hard to get. But this information exactly determines why a model succeeds or fails. Benchmarks are important, sure, but why did Cursor choose Kimi's model? Kimi will definitely think very seriously about this question, or just go ask them. This kind of experience regarding frontier AI real-world use is extremely valuable and hard to get. They know what kind of open-source model will be better. And this knowledge can only be gained by frequently releasing models. Nvidia is now also building momentum in releasing models, and will get more of this feedback later. The US's Reflection AI might say, "We will release when the model is ready," but then they don't get the feedback of real users using the model. "The benchmark is good, but it fails in this highly relevant scenario. Why?" So I feel that the intangible knowledge of how an open ecosystem operates is precipitating on the Chinese side.

Chen Qian: Conversely, what is the biggest disadvantage for Chinese labs today? Is it still GPUs?

Nathan Lambert: Yes, it's compute. These estimates might not be perfectly accurate, but you can roughly say that Meta or OpenAI possesses more GPUs than all these Chinese companies combined. And it's very likely true. I've heard rumors that a single OpenAI researcher can get about 5,000 GPUs. For a Chinese open-source model lab, that might be a very large chunk of their entire research budget. So their space to take risks and run dynamic algorithmic experiments is much narrower. They need to continuously optimize along the current model path, treating this as a necessity because they know this path can still improve. But the whole ecosystem is less dynamic. One Chinese lab told us their latest pre-training run took about six months, and that model is now released on Hugging Face. If that training run failed, this entire generation of models might not exist at all. For a company, this is a very dangerous position. I think under existing compute conditions, they have already achieved results beyond most people's expectations. But compute is indeed the limiting factor. Chinese labs genuinely crave more Nvidia GPUs.

Chen Qian: How much will compute scarcity change their model strategies?

Nathan Lambert: Most importantly, every model is basically designed based on your anticipated downstream usage and the shape of your training cluster. The training cluster determines the efficiency of training models of different sizes. At the same time, you know you have to serve this model. Because they might have smaller training clusters, this changes many decisions. It might also lead them to change fundamental model architectures less often, because pre-training runs longer. This is more obvious for smaller labs. Alibaba and ByteDance are somewhat special cases due to their scale of resources. But for MiniMax, Moonshot—models the outside world generally considers to be in the Claude and GPT class, probably in the 3 to 6 trillion parameter range—if they can't get the latest NVL72 racks, Blackwell chips, and such resources, it will be much harder for them to continue scaling up. So they will hit a ceiling. But this is again a multi-objective optimization problem. What is your training cluster? What is your inference stack? Do you hope to run on Huawei inference chips now? Who are your users? All these affect the design. What we finally see is just a 600 billion to 1 trillion parameter model on Hugging Face. The mapping behind it is very complex, but this is one of a lab's most interesting decisions, and most won't tell you the details.

Chen Qian: Will compute scarcity force higher efficiency, or will it ultimately limit the ceiling?

Nathan Lambert: Both. But more accurately, it changes the way you work. Everyone wants compute efficiency because if your research is more efficient and you halve the cost of an experiment, you can run twice as many experiments. This is very basic math that every AI researcher wants. It's hard to say Chinese labs are necessarily better at efficiency. Maybe because they have less inference compute, they train models that are more efficient at inference. But that's a different trade-off. DeepSeek is famous for some VRAM-saving innovations, some of which are now widely adopted. But from a US lab's perspective, this kind of optimization can sometimes seem like trying too hard. It yields some benefit, but might not be worth that much effort.

Chen Qian: How do domestic accelerators perform? Like Huawei chips. How are they in practice?

Nathan Lambert: I hear from many people that inference is basically usable, but training isn't yet. This is basically the consensus. We went to some less core companies—not core LLM companies—and they'd say, "Yes, we must have Huawei chips, but haven't used them much." But most of the language model companies we talked to are mature enough; they have services, people are registering and using them. For example, Zhipu's API processes trillions of tokens daily. So these companies all say, "If you can give me Huawei chips, we might be able to get more customers." They are figuring out how to use Huawei chips for inference. But for training, right now only very small models might be trained on Huawei chips. Personally, I expect this state to continue for a while. So researchers focused on training all say, "I want more Nvidia chips."

Chen Qian: How long is "a while"?

Nathan Lambert: Best to ask a semiconductor expert, but I expect this will take at least a few more years, because Nvidia is still so far ahead right now. He is mobilizing the global supply chain to make the top-tier product. Whereas on Huawei's side, it's more like demand forcing innovation. How far they can go, I don't know.

Chen Qian: What do you think of the recent debate between Jensen Huang and Dwarkesh? To brief those unfamiliar with the context: Jensen opposes strictly limiting China's access to Nvidia GPUs, arguing China already has the energy infrastructure, chip production capacity, and AI talent to independently develop advanced models. Furthermore, he says restricting sales only makes Nvidia lose a massive market without stopping China's technological progress. Dwarkesh, however, emphasizes the "China threat" theory more. Of course, Jensen has his own motives, as he wants to sell more chips in China. Who do you think makes more sense?

Nathan Lambert: Realistically, I'm in the middle, but leaning more towards Dwarkesh. Ultimately, this becomes a geopolitical discussion. Do you believe AI will be used for state power competition? As an American, I prefer traditional American values, like freedom of expression, spreading more globally. I also believe providing AI chips to China does indeed bring some risk to the balance of power. So as an American, I'd probably stand there. But the problem is the degree is hard to judge. If you listen to people inside Nvidia, they basically say the US doesn't have enough power to build more data centers, so nobody in the US can buy these chips. "We want to sell to China." But at the same time, US companies are demanding larger quotas. US policymakers also hear US companies say they would have bought more. So I've talked to both sides. US labs say, "I want more chips to grow my company." Nvidia's policy researchers say, "The US can't absorb more volume." It's hard to get the real data. How much will the US actually buy? And how many chips have been diverted to the Chinese market? That's why I have a lot of uncertainty on this issue. I don't know exactly where the ultimate balance point is. So maybe there is indeed a portion of chips Nvidia couldn't sell in the US anyway. Then maybe we should allow them to sell to China to support Nvidia maintaining its AI leadership. But I really can't say.

I'm more curious if the ban will stimulate China to build better AI chips, namely Huawei chips. This will definitely happen. We talked about the difference between SaaS and cloud earlier. I think AI will be more like cloud spending. This will support various new companies in China, because they need the cloud to build viable products, and will bring ever-growing inference demand. We talked about OpenClaw and Claude Code; these show the inference demand already exists. Adoption in China is even earlier. You talk to Chinese developers, they say, "I use Claude." That means this demand is currently being routed to Western compute. As China unlocks more domestic AI inference capability, the demand generated by these use cases will converge on one question: how many chips can Huawei make? So this isn't years away, but months away. New applications that massively empower developers, like Claude Code, will appear in China. People will have a very strong demand for them. So I definitely see this happening. From this angle, Jensen Huang is partly right: this will drive the development of a domestic AI tech stack. Domestic labs already have Huawei chips and are designing inference solutions for them. In a sense, this has already happened. The remaining questions are how big the scale is, and to what extent people deem it acceptable. So the bottleneck is really Huawei's production capacity.

Chen Qian: Right. And the ecosystem. Whether models can easily run on these chips—that is, what is the corresponding CUDA ecosystem like.

Nathan Lambert: I am very bullish on AI demand continuing to grow, so demand will create this necessity.

Chen Qian: Anthropic's CEO Dario Amodei has been one of the tech leaders most publicly warning that China's AI development poses national security and geopolitical threats. What do you think of his approach?

Nathan Lambert: I think it's a bit too aggressive. Some of Anthropic's media statements are actually quite careful to distinguish between concerns about the Chinese government and views on individual Chinese scientists. But Dario's expressions on short-term risks, especially the risks of open-weight models, have been somewhat overly intense. So by playing this role of the sharpest vocal critic on this issue, it might sound a bit uncomfortable for some people.

Chen Qian: You mentioned that China's weakness in the AI data industry is more important than people realize. Can you elaborate on that?

Nathan Lambert: One of the things that surprised me most was that China's data industry is nowhere near as developed as America's. In the current environment in the US, companies like Anthropic and OpenAI will say they have billions of dollars in data budgets to buy training environments from external vendors. We asked Chinese companies, and they said, "No, we basically don't have that kind of thing." Some do it internally. People mention ByteDance or Alibaba, companies large enough to build internal data teams. But almost no one says, "We go to external vendors to buy data." Whereas in the US, this is incredibly common. It's practically the standard playbook for frontier labs. Realistically, training a model sometimes means: whichever part of the model is weak, you go buy data for that part to see if you can pull the performance up. Of course, they also do more exciting research, but this boring data engineering is extremely important. And this dynamic doesn't seem to exist in China. This is the biggest difference I saw. OpenAI researchers obviously have more compute than Chinese researchers, so Chinese startups might work on different problems and look more efficient. But data doesn't inherently *have* to be different, yet it *is* very different.

Right now, the understanding of US frontier models is that they will become knowledge work tools. For a typical knowledge worker like me, I use Claude Code and Codex to handle most of my work now. People will see them enter medical, legal, consulting, financial fields, etc. And a lot of the training data

for these models in these fields is acquired through networks of data companies. They pay professionals to create training data. The US has a complete market: Scale AI, Mercor, Surge, Turing, and many other companies. The market is pushing them to shift from doing offshore preference data for RLHF to highly skilled labor industries. The scale is massive. The amounts model labs pay for this are in the billions of dollars. They do this because it's genuinely valuable; it makes models stronger at these agentic tasks. So the question is, can Chinese labs open up the domestic market for their own agents? If people truly use their models for these things, the demand will give them the ability to buy this expensive data. Conversely, if Chinese AI researchers and knowledge workers all say, "I just use Claude," then Claude has even more reason to keep doing these things in the US, and the data industry will remain in the US. I talk to friends doing post-training in labs, and they say these data companies are truly crucial for getting incremental performance gains across various domains. If China doesn't have such an industry, they must be very creative. You might have AI researchers construct a financial task environment themselves, like building an Excel spreadsheet. But that's not a task AI researchers are naturally good at. That's another world.

Chen Qian: I have a few more questions about Chinese demand and models. In the mobile internet era, Chinese companies were very good at launching new apps. People say companies like Meituan, WeChat, and Alibaba completely changed our daily lives. So there's an assumption: "Once the models are ready, China will run much faster on the AI application layer this time as well." Do you think that will happen?

Nathan Lambert: I think both the US and China will be extremely active on the application layer. Some might even say US consumers are more used to downloading new apps, using them, and paying for them, whereas China has these super apps. The question is, "Can super apps support the same diversity of new application scenarios?" In the US, you see products like Sora suddenly become number one, millions of people try it, and then it might fall off its pedestal. It has its own sandbox and can offer a more explicit experience. So I think on the software layer, the US ecosystem is better suited for rapid trial-and-error and iteration of new applications.

Chen Qian: Do Chinese labs worry about commercialization?

Nathan Lambert: They didn't express it very explicitly. Then again, we didn't ask. I also don't think many frontier AI labs are overly obsessed with commercialization. OpenAI and Anthropic want to go public, but most people building AI models will say, "I want to build the best model." If you believe models will get stronger and more useful, fundraising won't be an issue. It depends on who you ask, right? If you ask product managers, they might worry, or if you ask leadership who needs to keep buying more compute.

Chen Qian: Should US labs visit Chinese labs more often?

Nathan Lambert: I think some US companies, especially small and medium players creating economic value around open-source models—whether doing fine-tuning APIs, specialized deployments, etc.—will want to build relationships with Chinese labs. Because Chinese labs are very willing to share technology and experience. After going to China, many companies asked me, "How did you get in touch with those labs? We want to go too." If you look at US companies building infrastructure or training models, they are all built on top of open-source models. If this trend continues, we will enter a new era where these new companies will start learning cross-border collaboration. I expect more US companies to go to China because getting a visa is much easier now. I used a transit visa exemption (entering and exiting to different countries). It was very simple; basically, you just show up. In contrast, Chinese researchers or engineers going to the US have to wait for the visa process. So I expect more US mid-sized startups

interested in open-source models to go to China. It should happen in the next few weeks to months. These things happen fast. Summer 2026, it's very interesting.

Chen Qian: You are a frequent guest on Lex Fridman's podcast. He also recently went to China. He has a video of riding in a truck that went viral on the Chinese internet. Have you and Lex talked privately about your China trips, or did he mention wanting to go to China?

Nathan Lambert: We haven't done that kind of "China debrief" conversation yet. I think we are both just interested in this massive population of humanity that profoundly influences the technology we live with. And seeing it with your own eyes is the best way to understand things. Of course, Lex travels much more than I do; he goes many places. I can't speak for him, but I guess there are some similarities in motivation.

Chen Qian: Great. Can you make some predictions? 12 months from now, what will happen to Chinese labs and US labs respectively?

Nathan Lambert: I think next, some new kind of agent product will emerge and become the thing everyone talks about. Claude Code and Codex will still be there, but there will be other products, maybe from Google, maybe from someone else. I think we are still at the very early stages of complex agent workflows. So that will be a new thing. What's unpredictable is whether models will just keep getting better the way they are now, or if there will be a lucky breakthrough in areas like continuous learning. I'm more inclined to think we will see revenues for AI labs, big and small, continue to grow. OpenAI and Anthropic might go public and be fine. But what's more interesting is the mid-sized companies. For example, Together, Fireworks, Baseten, and some Chinese labs like Kimi, Zhipu, and MiniMax. They aren't massive, but they will sell tokens, sell models, sell various intelligent services. They will do very well in "selling intelligence," because demand is so high. I expect the generally accepted gap between US and Chinese frontier models will widen. People used to say three months, later they might say six to nine months, or a year. But Chinese labs don't seem like they will stagnate or give up. We will very likely see models like DeepSeek V5, Kimi K3, GLM 6, etc. 12 months is still a very short time. We are still seeing demand rising rapidly. The market will likely still be crazy. Or we haven't reacted yet; we haven't fully digested this cycle of demand. So it might just seem like a continuation of current trends. If you ask about three to five years, I think some major Chinese lab might fail due to funding issues. I don't know, say companies like MiniMax or Zhipu. The US market might also have more corrections. At the three to five, or ten-year scale, things will be different.

Chen Qian: After this trip, are you more optimistic or more worried about the global AI race?

Nathan Lambert: I'm probably fairly neutral. My main worry about AI is focused on US public opinion potentially turning more negative towards AI, which would hurt progress. The global AI race largely aligns with my geopolitical expectations: in the biggest picture, the US leads, and China wants to catch up. What's truly complex is how many chips are sold, and how these chips will ultimately affect the ecosystem. I think the main thing is to bring more humanity to it, doing my best to make both sides feel less extreme. Many of the people building and using the technology have almost no say in top-level decisions. A small part I can play is helping both sides understand each other better.

Chen Qian: Thank you very much, Nathan. Those are all my questions. Is there anything else you'd like to add, or any questions I missed?

Nathan Lambert: No, thanks for having me. I hope to visit China again soon; there are still many places worth seeing.

Chen Qian: Looking forward to doing our second interview after your next trip to China.

Nathan Lambert: Thank you.

Chen Qian: Okay, thanks.

(Note: The following section appears to be a tour of the Ai2 office, transitioning from the main interview.)

Chen Qian: Tell us what you guys do here.

Nathan Lambert: Ai2 does all kinds of open-source AI research and development. It used to be very academic, but over time, it's increasingly shifted towards this "artifact-centric" development. You build models, codebases, and datasets, letting the whole community continue building on top of them. OLMo and these open language models are among the most popular, but there's also the Molmo multimodal model, OlmoEarth, etc. Additionally, Ai2 has historically been well-known for "integration with academic literature." The application for this is Semantic Scholar, which is somewhat seen as an alternative to Google Scholar for retrieving academic literature. This team later evolved into building AI agents for science, under the brand Asta. They are building tools to help people construct and understand scientific literature. This is the core of Ai2's deepest culture. It's so "scientific" and holds a community value set, bringing many tangible benefits to the research community. That's really great. This trait is reflected in those meticulously crafted, highly detailed papers. We have so many directions we want to go. Which way do you want to go? There are too many different directions. One of the best things about working at Ai2 is that you are fully absorbed in building things that are truly used by various researchers. When you travel around as a scholar, you find people everywhere building on top of OLMo and knowing Ai2. This is an impact that will last a long time. It's really nice. Plus, we have interns and student researchers who go into the academic community, into various other research institutions, and spread these scientific values. This is a very meaningful thing Ai2 does.

Chen Qian: How many people do you have now?

Nathan Lambert: Around 200, I think.

Chen Qian: Wow, quite a few. And most of them are researchers, right?

Nathan Lambert: I'd estimate about half the talent is technical, doing engineering on the traditional AI research side. Plus there are engineering teams supporting our compute cluster and some products.

Chen Qian: How is a non-profit lab different from other labs in Silicon Valley?

Nathan Lambert: Ai2 is a pure non-profit. This means most of its funding comes from philanthropic donations. The name "Allen" is on the door, from Paul Allen, one of Microsoft's co-founders. It also gets some large grants, like from the National Science Foundation. So it operates entirely on non-profit funding and basically doesn't sell any products. It just shares everything it builds for free for everyone to learn from.

Chen Qian: I guess you have more research freedom.

Nathan Lambert: Yes. On the spectrum from closed labs to academia, resource-wise we are in the middle. Freedom-wise, we have a bit more resources. But there is also pressure; we must build models that are truly useful. So that does suppress freedom a little bit, but nowhere near as much as a frontier lab sprinting towards AGI. However, if you walk into a true academic institution, you'll see a diversity and spark of ideas that you can't get in a company. Companies can't bypass "making money to pay the bills."

Chen Qian: How is the vibe in Seattle different from Silicon Valley?

Nathan Lambert: It's much calmer here. There aren't as many parties or events. I actually wish there were more people in Seattle. There are several major tech talent hubs here, like Amazon and Microsoft; the big tech companies should get together more. But that's just how it is. In some ways, I think it's better for settling down and starting a family because it's more spacious and life isn't as tense. The overall vibe is pretty good. Here, you don't talk about AI everywhere you go. Whereas in the Bay Area, it feels like AI is the *only* topic right now. I understand why—I buy into it myself.

Chen Qian: But you also go to the Bay Area often.

Nathan Lambert: I go regularly, maybe three to eight times a year. I studied at UC Berkeley, so I lived in the Bay Area for seven or eight years. And you have to go back every now and then to keep up with what's really happening there. But in the Bay Area, you don't have this kind of scenery. Such a beautiful lake. What's it called?

Chen Qian: Lake Union. This is absolutely my dream office. When you went to China, what did you think of the offices there? The AI lab offices.

Nathan Lambert: They didn't have this view. But they are also tech company offices, so they are quite similar in that regard. Walking in, I felt like I was in a tech company. I've been in a lot of tech company offices in my life.

Chen Qian: The "showroom culture" is different.

Nathan Lambert: Oh right, the showrooms. That's very unique. That's a very Chinese thing. Most people probably have to see it for themselves the first time, and then they realize, "Oh, this is really a thing."

Chen Qian: Great. Thank you so much, Nathan.

Nathan Lambert: Thanks for having us.

Chen Qian: You're welcome. Thanks for coming.

(Video outro)

Chen Qian: That was my conversation with Nathan, as well as the Ai2 office tour. Thanks for watching. Your comments, shares, and likes are the best motivation for us at Silicon Valley 101 to produce deep tech and business content. I'm Chen Qian. We'll see you in the next video! Bye.